

ACT 1 — what a filter is: one real detection from camera C, smoke1_20260716

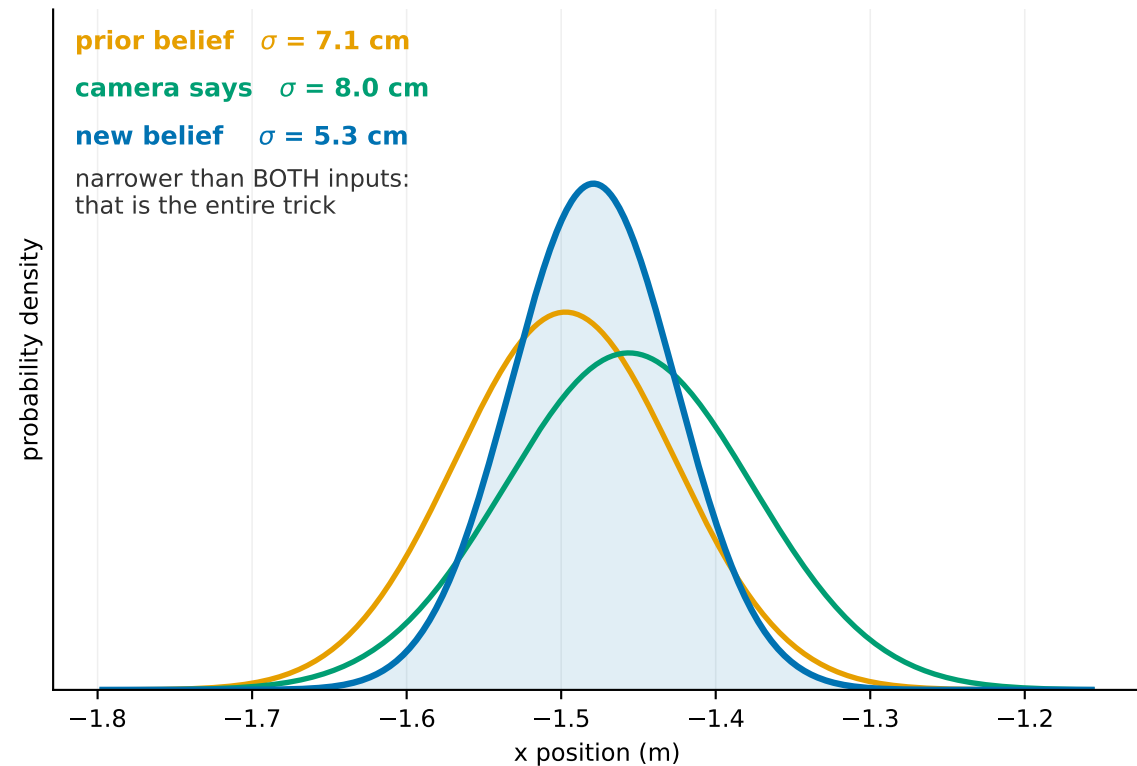
A filter multiplies two Gaussians
the answer is always narrower than either

prior belief $\sigma = 7.1$ cm

camera says $\sigma = 8.0$ cm

new belief $\sigma = 5.3$ cm

narrower than BOTH inputs:
that is the entire trick



The same update in 2-D
ellipses are the stated 95 % regions

